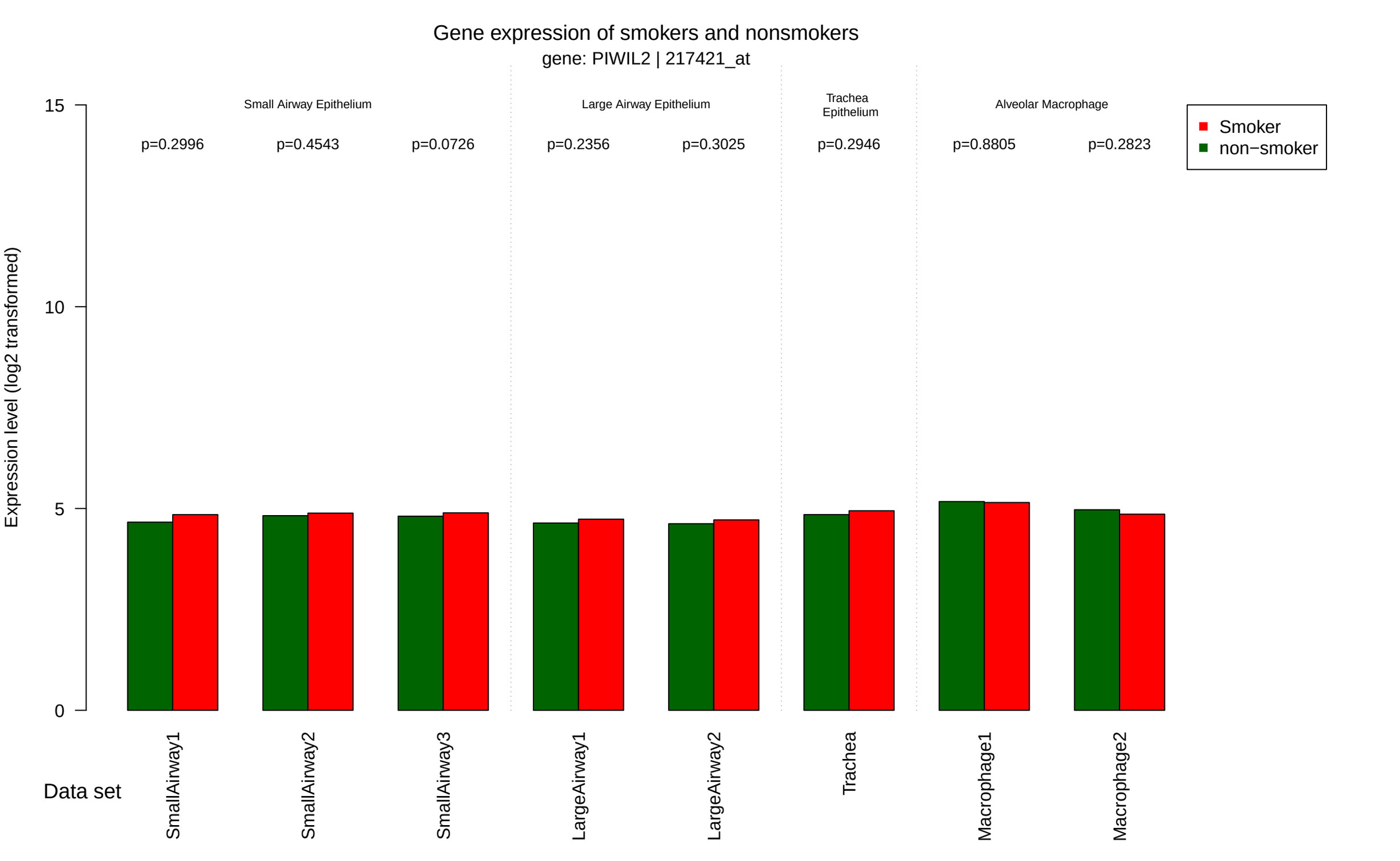
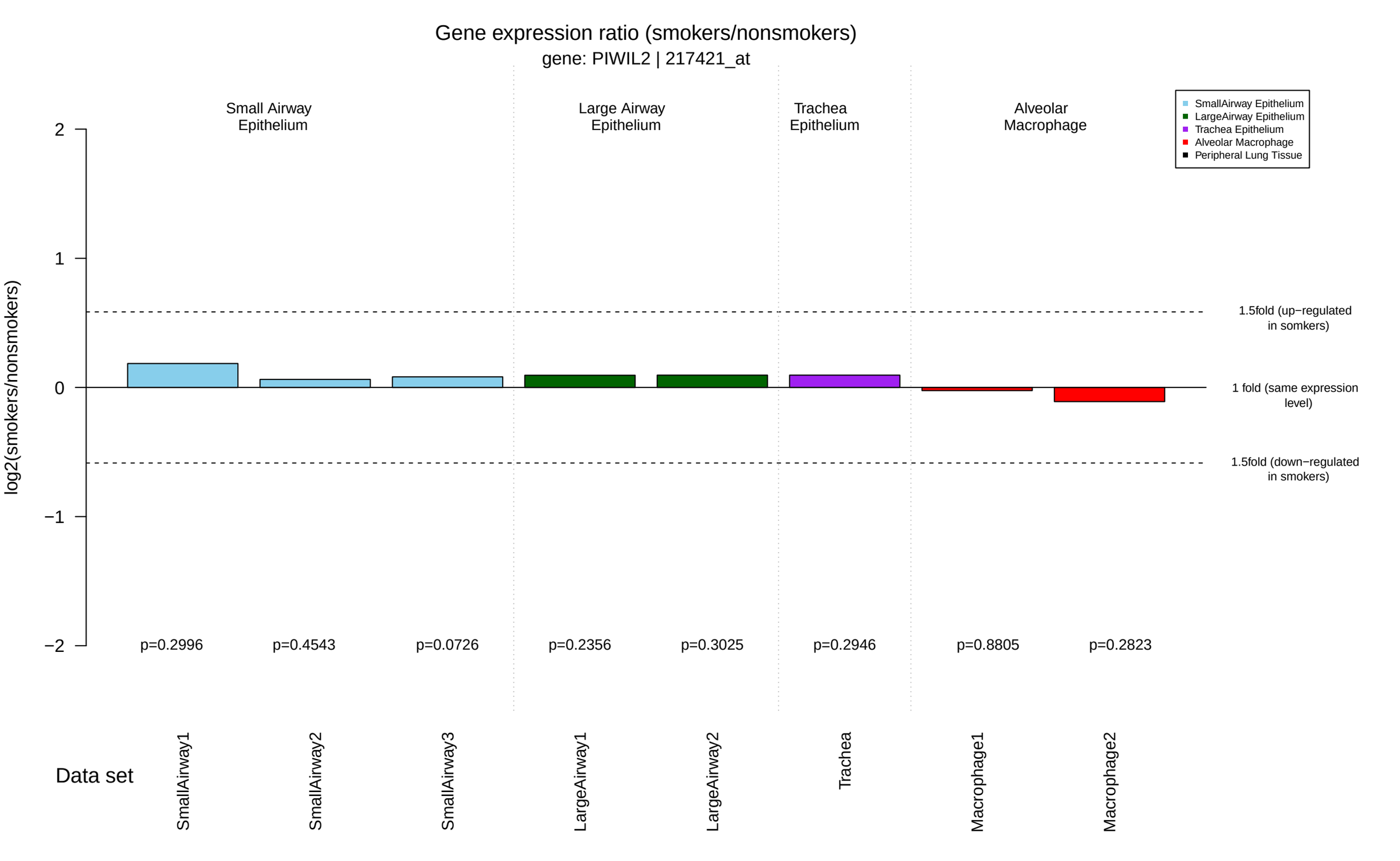
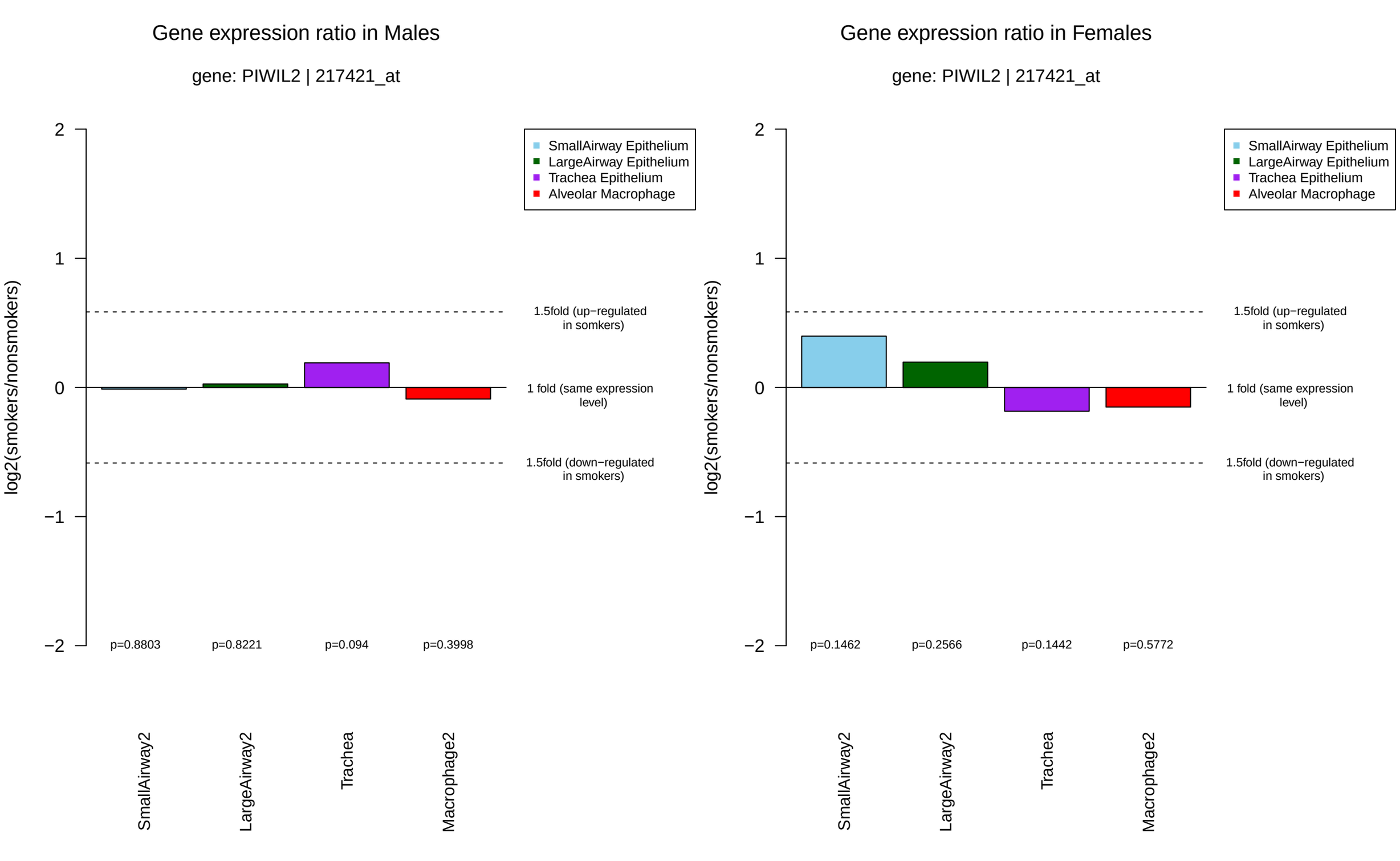
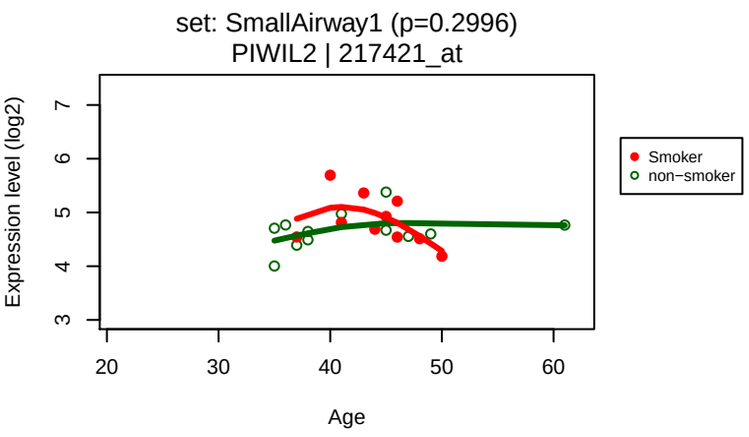
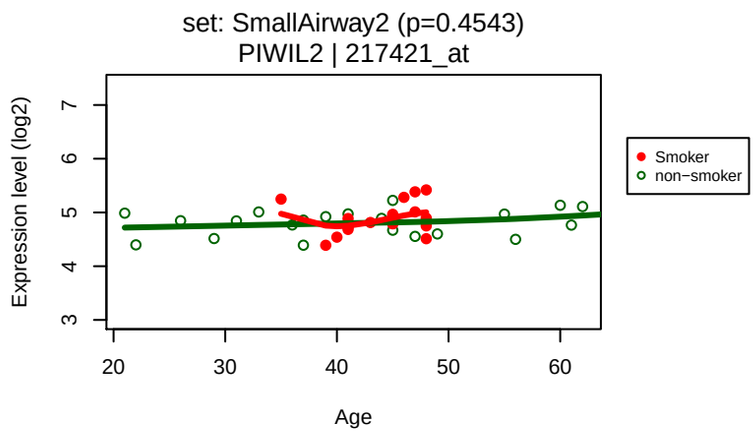
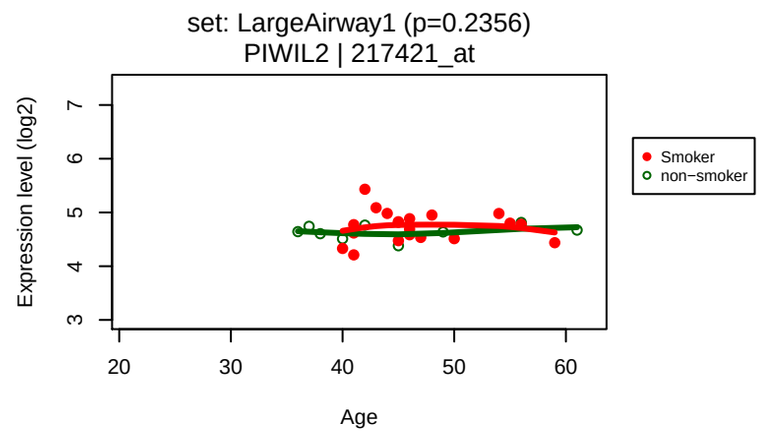

Gene expression of smokers and nonsmokers
gene: PIWIL2 | 217421_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2996
p=0.4543
p=0.0726
p=0.2356
p=0.3025
p=0.2946
p=0.8805
p=0.2823
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

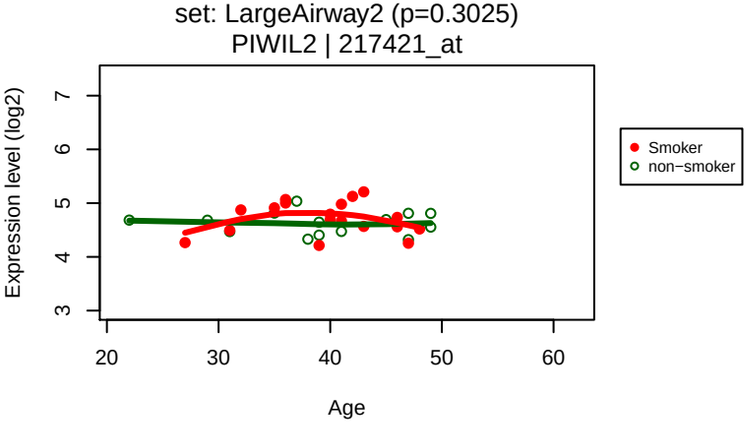

Gene expression ratio (smokers/nonsmokers)
gene: PIWIL2 | 217421_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2996
p=0.4543
p=0.0726
p=0.2356
p=0.3025
p=0.2946
p=0.8805
p=0.2823
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

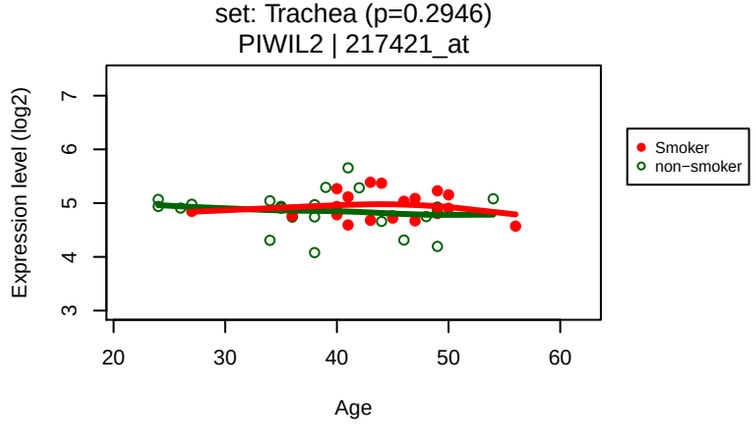

Gene expression ratio in Males
gene: PIWIL2 | 217421_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8803
p=0.8221
p=0.094
p=0.3998
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIWIL2 | 217421_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1462
p=0.2566
p=0.1442
p=0.5772
SmallAirway2
LargeAirway2
Trachea
Macrophage2

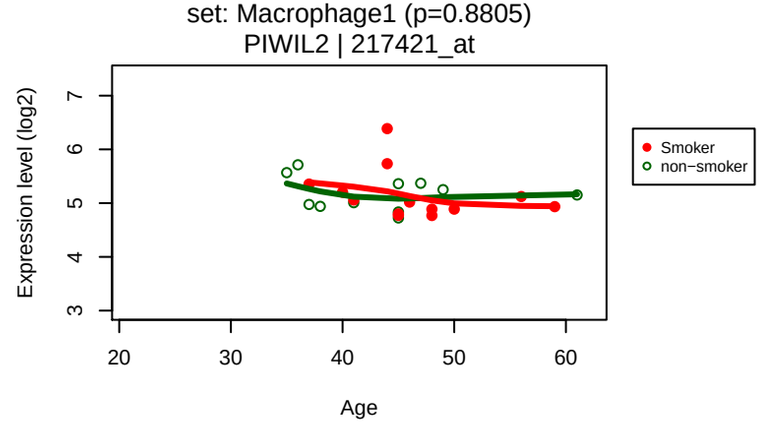

set: SmallAirway1 (p=0.2996)
PIWIL2 | 217421_at
Expression level (log2)
Age
Smoker
non−smoker

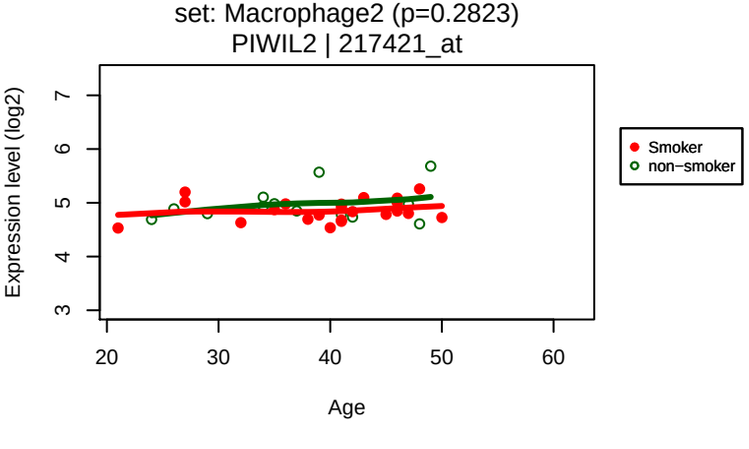

set: SmallAirway2 (p=0.4543)
PIWIL2 | 217421_at
Expression level (log2)
Age
Smoker
non−smoker

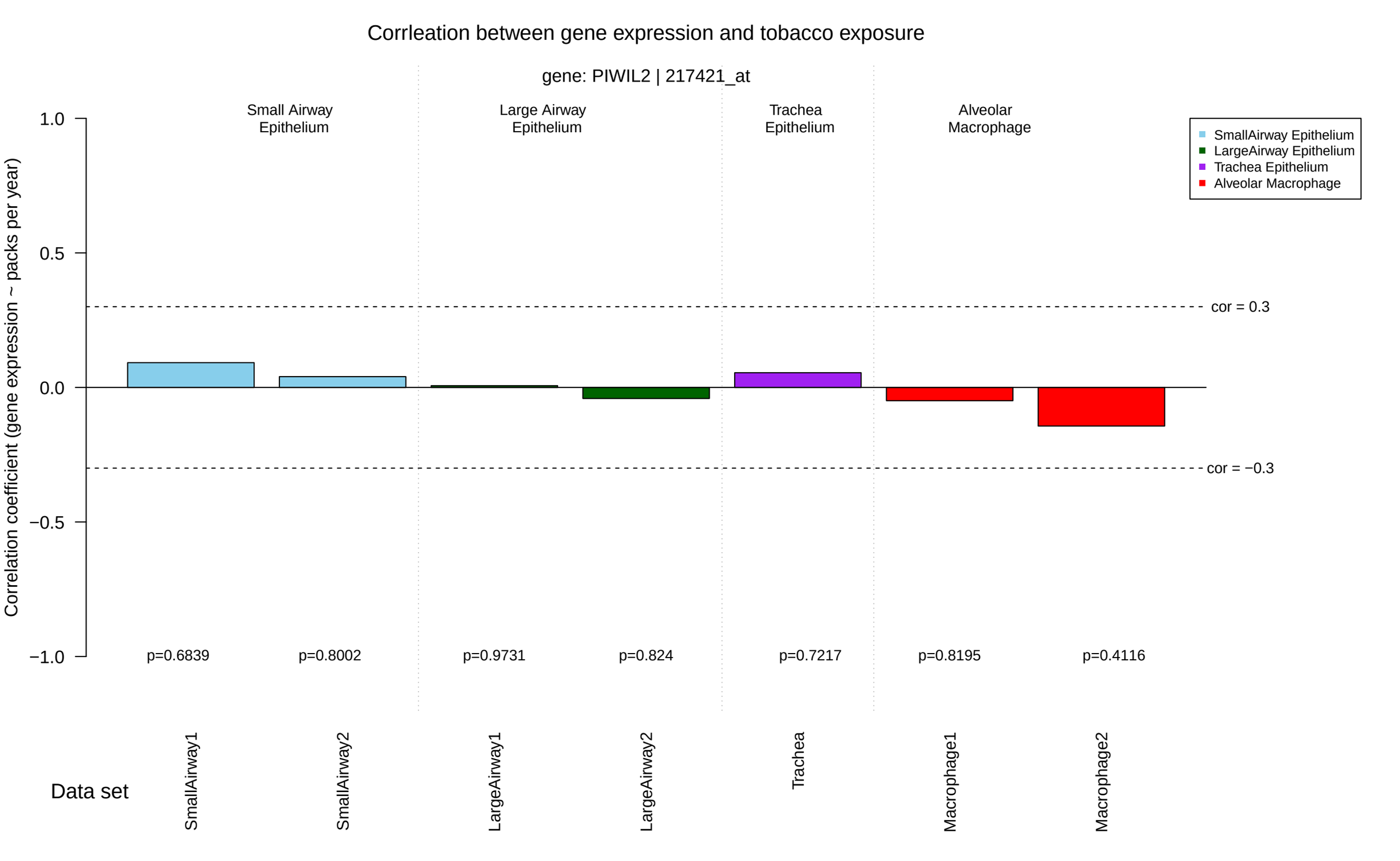

set: LargeAirway1 (p=0.2356)
PIWIL2 | 217421_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3025)
PIWIL2 | 217421_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2946)
PIWIL2 | 217421_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8805)
PIWIL2 | 217421_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2823)
PIWIL2 | 217421_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIWIL2 | 217421_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6839
p=0.8002
p=0.9731
p=0.824
p=0.7217
p=0.8195
p=0.4116
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2